

Soham Patel

Software Engineer — Industrial AI, Digital Twins & Computer Vision

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PROFILE

Software engineer with 2.5 years of production experience in regulated healthcare data and payment systems, now specialising in industrial AI. I build backends that are tested, containerised and observable — FastAPI and .NET services, Azure ETL, CI/CD — and apply the same discipline to machine learning: a browser-native digital twin of a fischertechnik Industry 4.0 cell driven by its real PLC process image over MQTT and OPC-UA; YOLOv8 exported to ONNX for on-device inference; LIDAR SLAM with occupancy-grid mapping and pure-pursuit control; latent diffusion for medical image anomaly detection. M.Sc. Information Technology as Gold Medalist, Rank 1 (German equivalent 1.2); currently completing an M.Sc. in Artificial Intelligence for Industrial Applications at OTH Amberg-Weiden.

PROFESSIONAL EXPERIENCE

03/2025 – 08/2025

Part-time

Teaching Assistant — Decision Modelling & Data Analysis

OTH Amberg-Weiden · Weiden, Germany · Supervisor: Prof. Dr. Simon J. Preis

- Designed and delivered the applied data-analysis strand of a Master's module to a cohort of 30: SQL, statistical modelling, classification, regression and model evaluation.
- Built and maintained the KNIME reference workflows used across the module, covering data cleaning, feature engineering and evaluation on real industrial datasets.

02/2023 – 02/2024

Full-time

Software Engineer — Data Systems

Altera Digital Health (Allscripts India LLP) · Pune, India / Remote

- Architected high-throughput ETL pipelines on Microsoft Azure for healthcare audit-log data, reducing data-retrieval latency by 30% on compliance-grade patient datasets.
- Brought the pipeline to full compliance with HIPAA-aligned data-handling standards — encryption in transit and at rest, access auditing, and retention rules enforced in code rather than by convention.
- Refactored legacy C#/.NET services and introduced a pytest suite that gates every deployment, cutting technical debt by an estimated 25% with zero production incidents over the period.

C# · .NET · Microsoft Azure · Blob Storage · T-SQL / MSSQL · Azure DevOps · pytest · Git

08/2022 – 01/2023

Full-time

Python Backend Engineer

Electrum IT Solutions · Vadodara, India

- Designed and ran the REST API layer of a fintech platform processing 2,000+ transactions per day at 99.9% uptime, in FastAPI and Django.
- Implemented JWT authentication and role-based access control across the service boundary, and multithreaded the settlement path to hold response times flat under load.
- Introduced centralised payment-failure logging and fault isolation, reducing mean time to diagnosis for production incidents by roughly 60%.

Python 3 · FastAPI · Django · PostgreSQL · JWT · REST · Linux

12/2021 – 04/2022

Internship

Full-Stack Engineering Intern

Promact Infotech Pvt. Ltd. · Vadodara, India

- Delivered an expense-management product end to end: Angular 8 front end, ASP.NET Core 5 API, 9-table normalised MSSQL schema, JWT auth with role-based access, WebSocket live balances.
- Designed a graph-theory debt-simplification algorithm reducing the transfers needed to settle a group by up to 33%.

Angular 8 · ASP.NET Core · MSSQL · WebSocket · JWT · Agile

EDUCATION

10/2024 – present

M.Sc. Artificial Intelligence for Industrial Applications

Ostbayerische Technische Hochschule Amberg-Weiden · Amberg, Germany

- Current average 2.0 · 45 ECTS completed. Focus: digital twins and Industry 4.0, computer vision, deep learning, NLP, autonomous robotics, modern databases (Embedded Intelligence graded 1.3).
- Thesis direction: physics-informed latent world models as adaptive digital twins for sample-efficient robot skill transfer in contact-rich industrial assembly. Start August 2026.

08/2020 – 05/2022	M.Sc. Information Technology Charutar Vidya Mandal University · Gujarat, India – CGPA 9.55/10 — German equivalent 1.2. Gold Medalist, Rank 1 of a graduating class of approximately 120. – Software engineering, databases and distributed systems; thesis in applied machine learning.
06/2017 – 05/2020	B.Sc. Computer Science Sardar Patel University · Gujarat, India – CGPA 8.5/10 — German equivalent 1.5. Algorithms and data structures, operating systems, networks.

SELECTED PROJECTS

2025 – 2026	Industry 4.0 Digital Twin — fischertechnik Training Factory – Browser-native 3D twin modelled on the real 15 mm fischertechnik construction grid and driven by the actual PLC process image — 9 digital inputs and 14 digital outputs taken from the Belegungsplan — over MQTT and OPC-UA, with a predictive-maintenance dashboard on live sensor streams. <i>Python · FastAPI · MQTT · OPC-UA · InfluxDB · Three.js · Docker Compose</i>
2025 – 2026	Unsupervised Brain-MRI Anomaly Detection via Latent Diffusion – BraTS 2021 FLAIR slices → frozen MONAI AutoencoderKL ($32 \times 32 \times 4$ latent) → Diffusion-ModelUNet trained only on healthy anatomy → DDIM inference with partial noising at an SNR-selected T_{int}, with an ablation over T_{int} and the ELBO derived from first principles. – Result: DICE 0.076 — well below target. Reported as a negative result: reconstruction-error anomaly detection does not separate diffuse FLAIR hyperintensity from normal anatomical variation at this latent resolution. <i>PyTorch · MONAI Generative · HuggingFace · CUDA 12 · Docker</i>
2025	Autonomous Maze Navigation — LIDAR SLAM – Log-odds occupancy-grid mapping, frontier exploration, A* planning on an inflated obstacle radius, pure-pursuit steering; ported from a Python Webots controller into the browser. <i>Python · Webots · NumPy · Three.js · TypeScript</i>
2025	Real-Time Scene Understanding – YOLOv8n-seg exported to ONNX and run under ONNX Runtime Web, so camera frames never leave the visitor's device; paired with a self-hosted vision-language model for narration, with a detector-only fallback when none is reachable. <i>YOLOv8 · ONNX Runtime Web · TypeScript · FastAPI · Docker</i>

TECHNICAL SKILLS

Languages	Python 3 · C# / .NET · TypeScript · SQL (T-SQL, MSSQL, PostgreSQL, MySQL) · C++
Backend & Cloud	FastAPI · Django · ASP.NET Core · REST · JWT · Microsoft Azure · Docker · CI/CD · Git · Linux · pytest
ML & Vision	PyTorch · Scikit-Learn · XGBoost · MONAI · HuggingFace · YOLOv8 · ONNX Runtime · Diffusion models (DDPM/DDIM) · Anomaly detection · Explainability
Industrial & Data	Digital twins · MQTT · OPC-UA · PLC process images · InfluxDB · ETL pipelines · SCADA time-series · KNIME · Power BI
Robotics & 3D	Webots · SLAM (occupancy grids) · A* / pure pursuit · MuJoCo · robosuite · Three.js

AWARDS, LANGUAGES & ADDITIONAL

2025	Grade 1.0 (highest distinction) — keynote, OTH AI Research Conference: <i>Brownian Bridge Diffusion Models for 3D Shape-to-Image Medical Image Synthesis</i> .
2022	Gold Medal & University Rank 1 — M.Sc. IT, CVM University (CGPA 9.55/10).
Languages	English C1 (IELTS 6.5) · German A2, in active study toward B1 · Gujarati native · Hindi fluent.
Work authorisation	Valid German residence permit · EU Blue Card eligible via a recognised M.Sc. · no sponsorship required · available from September 2026.
Other	Department student representative (two years) · Football: university team captain.